Programming Assignment 1- Bayesian Inference, Monte Carlo Simulations, and Poisson

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Made in LaTex - TexStudio

Abstract

This report analyzes a manufacturing scenario with three machines producing items at different defect rates for the first 3 parts of it...A,B,C. Exact Bayesian calculations are performed to obtain posterior probabilities of machine origin given defective or non-defective items. THen, a Monte Carlo simulation with $N = 100{,}000$ samples validates the theoretical results. We examine the Convergence of the simulation as a function of sample size.

Finally, the analysis is extended to a Poisson model for flaw counts in this case: For two machines, and the posterior probability of machine origin given a specific number of flaws is computed both analytically and via simulation.

1 Initial conditions

A factory uses three machines (A, B, C) with production shares of 20%, 50%, and 30%, respectively. Their defect rates are 10%, 2%, and 5%.:

1. Exact Bayesian calculation of the probability that a defective (or non-defective) item came from each machine.
2. Monte Carlo simulation to estimate these probabilities.
3. Analysis of convergence of the Monte Carlo estimate with increasing sample size.

All computations were implemented in Python using NumPy, Matplotlib.

2 Part A: Exact Bayesian Calculation

2.1 Total Probability of Defect

Let $P(A) = 0.20$, $P(B) = 0.50$, $P(C) = 0.30$ be the prior probabilities. The defect rates are:

$$P(\text{defect}|A) = 0.10, \quad P(\text{defect}|B) = 0.02, \quad P(\text{defect}|C) = 0.05.$$

By the law of total probability,

$$\begin{aligned} P(\text{defect}) &= P(\text{defect}|A)P(A) + P(\text{defect}|B)P(B) + P(\text{defect}|C)P(C) \\ &= 0.10 \times 0.20 + 0.02 \times 0.50 + 0.05 \times 0.30 = 0.045. \end{aligned}$$

2.2 Posterior Probabilities Given Defect (Bayes)

Applying Bayes' theorem:

$$P(A|\text{defect}) = \frac{P(\text{defect}|A)P(A)}{P(\text{defect})} = \frac{0.10 \times 0.20}{0.045} \approx 0.4444.$$

Similarly,

$$P(B|\text{defect}) = \frac{0.02 \times 0.50}{0.045} \approx 0.2222, \qquad P(C|\text{defect}) = \frac{0.05 \times 0.30}{0.045} \approx 0.3333.$$

These have to sum to 1.

2.3 Posterior Probabilities Given Non-Defect

The probability of a non-defective item is $P(\text{non-defect}) = 1 - 0.045 = 0.955$. The conditional probabilities of non-defect are 0.90, 0.98, and 0.95 for A, B, C, respectively. Then:

$$\begin{aligned} P(A|\text{non-defect}) &= \frac{0.90 \times 0.20}{0.955} \approx 0.1885, \\ P(B|\text{non-defect}) &= \frac{0.98 \times 0.50}{0.955} \approx 0.5131, \\ P(C|\text{non-defect}) &= \frac{0.95 \times 0.30}{0.955} \approx 0.2984. \end{aligned}$$

All these values are extracted from the code.

2.4 Interpretation

Machine B produces half of all items but has the lowest defect rate (2%). Consequently, when a defective item is observed, it is less likely to have come from B ($P(B|\text{defect}) \approx 22.2\%$) than from A (44.4%) or C (33.3%).

3 Part B: Monte Carlo Simulation

A Monte Carlo simulation with $N = 100{,}000$ items was performed. Each item was assigned a machine according to the prior probabilities, and then its defect status was drawn from the corresponding defect rate. The fractions of defective and non-defective items originating from each machine were computed.

3.1 Simulation Results

	$P(\cdot\|\text{defect})$	$P(\cdot\|\text{non-defect})$
Machine A (theoretical)	0.4444	0.1885
Machine A (simulated)	0.455	0.185
Machine B (theoretical)	0.2222	0.5131
Machine B (simulated)	0.223	0.517
Machine C (theoretical)	0.3333	0.2984
Machine C (simulated)	0.322	0.298

Table 1: Comparison of theoretical and simulated posterior probabilities.

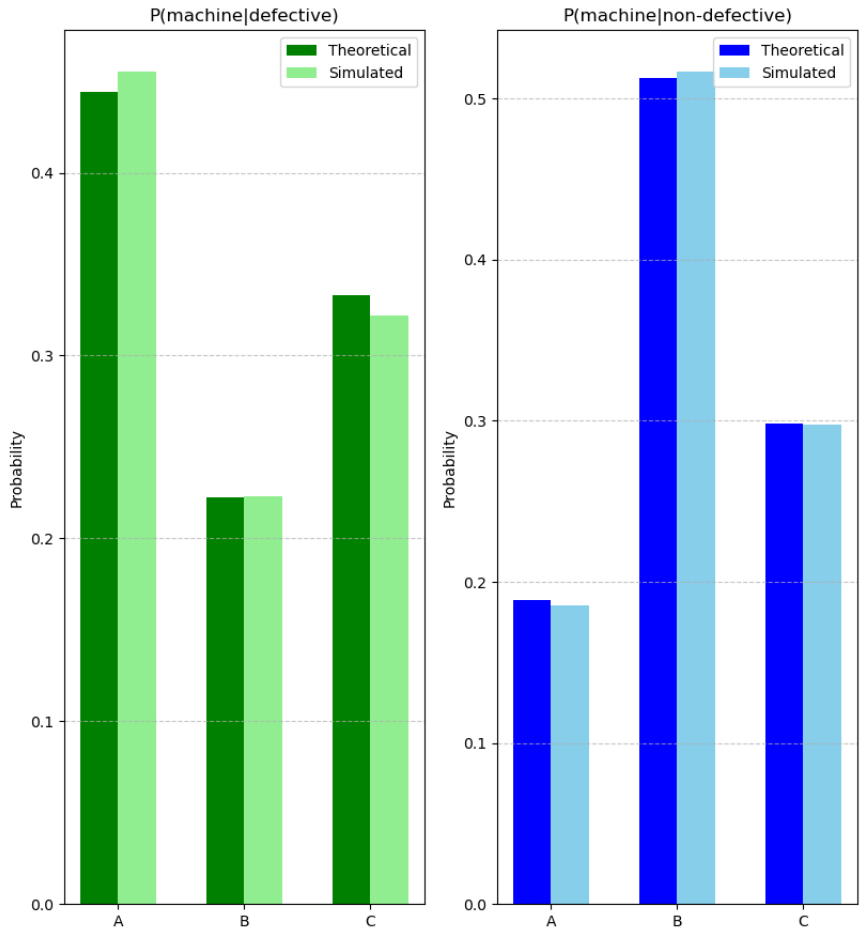

Figure 1: Theoretical vs. simulated probabilities given defect (left) and non-defect (right). Its very close due to the large N, play with the code I provided :)

4 Part C: Convergence as a Function of N

To study convergence, the Monte Carlo estimate of $P(A|\text{defect})$ was computed for sample sizes $N \in \{100, 500, 1000, 5000, 20000, 100000\}$. The absolute error from the exact value (0.4444) was recorded.

4.1 Convergence Plot

Figure 2 displays the estimate versus N on a logarithmic scale. As N increases, the estimate stabilizes around the theoretical value.

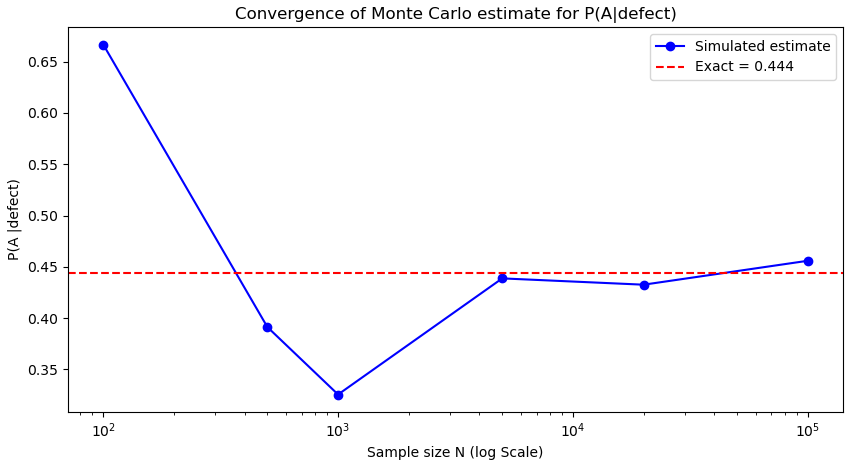

Figure 2: Convergence of Monte Carlo estimate for $P(A|\text{defect})$. The dashed red line marks the exact probability.

4.2 Error Analysis

The target absolute error was set to 1% of the exact value, i.e., 0.004444. For $N = 100{,}000$, the absolute error was approximately 0.0115, well above the threshold. None of the tested N values achieved an error below 1%. However, if you play with the prvided code you can reach that threshold.

5 Part D: Extension to the Poisson Distribution

Now suppose two machines (A and B) produce items with Poisson-distributed flaw counts. Machine A accounts for 40% of production with mean flaws $\mu_A = 3$; machine B accounts for 60% with $\mu_B = 1$. The probability mass function is

$$P(k|\mu) = \frac{\mu^k e^{-\mu}}{k!}.$$

5.1 Bayesian Posterior for Flaw Count k

Given an item with k flaws, the posterior probability that it came from machine A is

$$P(A|k) = \frac{P(k|A)P(A)}{P(k|A)P(A) + P(k|B)P(B)}.$$

Figure 3 shows $P(A|k)$ and $P(B|k)$ for $k = 0, \ldots, 5$. As k increases, the evidence strongly favors machine A because of its higher mean flaw rate.

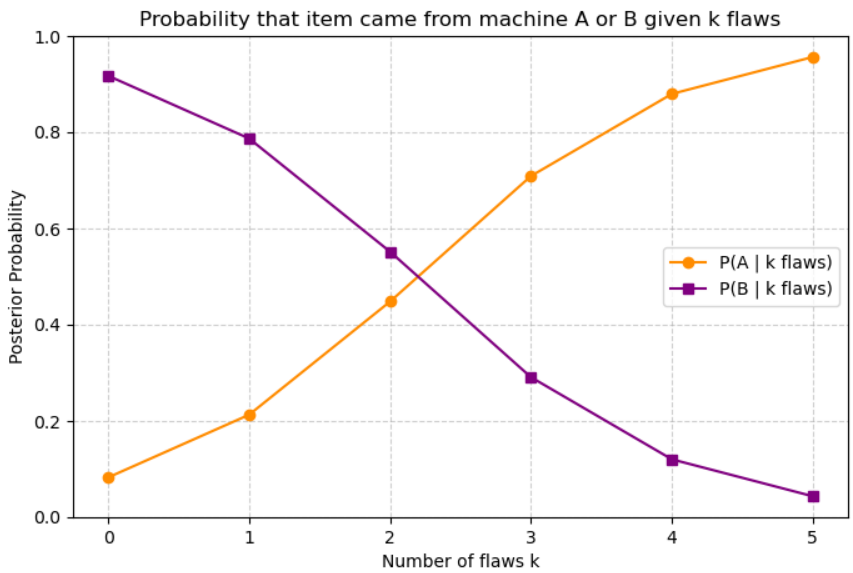

Figure 3: Posterior probability of machine origin given k flaws.

5.2 Monte Carlo Verification for $k = 2$

A simulation with $N = 200{,}000$ items was run.(Using the scipy library for the Poisson distribution) The analytical value for $k = 2$ is

$$P(A|k = 2) = \frac{\frac{3^2 e^{-3}}{2!} \cdot 0.4}{\frac{3^2 e^{-3}}{2!} \cdot 0.4 + \frac{1^2 e^{-1}}{2!} \cdot 0.6} \approx 0.4481.$$

The simulation yielded 0.449 with an absolute error of 0.001, confirming the analytical result.

6 Conclusion

Exact Bayesian calculations provide a clear interpretation of manufacturing defect data. Monte Carlo simulation accurately reproduces these probabilities and demonstrates convergence with increasing sample size. The Poisson extension illustrates how additional information (flaw counts) can dramatically update posterior beliefs. All code and figures are available for reproducibility ignore the seed. P.S. Never leaving Latex code for the last day.

Sources

- Python Graph Gallery. *Matplotlib examples and reference.* https://python-graph-gallery.com
- Cowan, Glen. *Statistical Data Analysis.* Oxford University Press, 1998.
- AI Assistance: DeepSeek was used for debugging complex Monte Carlo loop structures and Poisson likelihood derivations, specially Poisson.

- Special Thanks to Guido Gornati, without his assistance this report would have been impossible to write in time.